Lingjun Liu

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EDUCATION

North Carolina State University (NCSU) — Raleigh, NC, USA
Ph.D. in Computer Science | GPA: 4.0/4.0 | Advisor: Prof. Tim Menzies — Aug 2024 – May 2028 (Expected)

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
M.S. in Computer Science | GPA: 3.53/4.3 | Advisor: Prof. Doo-Hwan Bae — Sep 2019 – Aug 2021
Exchange Program in Computer Science — Aug 2018 – Jun 2019

National Tsing Hua University — Hsinchu, Taiwan
B.S. in Computer Science | GPA: 3.75/4.3 — Sep 2014 – Jun 2019

PUBLICATIONS

- **Learning Compiler Fuzzing Mutators from Historical Bugs** [MSR]
  Lingjun Liu, Feiran Qin, Owolabi Legunsen, Marcelo d'Amorim
  International Mining Software Repositories Conference (MSR), 2026
- **Development of Reliability Measurement Method and Tool for Nuclear Power Plant Safety Software**
  Lingjun Liu, Wooyoung Choi, Eunkyoung Jee, Duksan Ryu [TKIPS]
  The Transactions of the Korea Information Processing Society, 2024
  – Extended version of the KCSE 2024 paper
- **A Reliability Evaluation Tool for Nuclear Power Plant Safety Software**
  Lingjun Liu, Wooyoung Choi, Eunkyoung Jee, Duksan Ryu
  Korea Conference on Software Engineering (KCSE), 2024
  Best Short Paper Award 🏆
- **Search-based Test Case Selection for PLC Systems using Functional Block Diagram Programs** [ISSRE]
  Miriam Ugarte Querejeta, Eunkyoung Jee, **Lingjun Liu**, Pablo Valle, Aitor Arrieta, Miren Illarramendi Rezabal
  International Symposium on Software Reliability Engineering, 2023
- **MuFBDTester: A Mutation-Based Test Sequence Generator for FBD Programs Implementing Nuclear Power Plant Software** [STVR]
  Lingjun Liu, Eunkyoung Jee, Doo-Hwan Bae
  Software Testing, Verification and Reliability, 2022
- **An Empirical Study of Reliability Analysis for Platooning System-of-Systems** [QRS-C]
  Sangwon Hyun, **Lingjun Liu**, Hansu Kim, Esther Cho, Doo-Hwan Bae
  International Conference on Software Quality, Reliability and Security Companion, 2021
- **Attack-driven Test Case Generation Approach using Model-checking Technique for Collaborating Systems**
  Zelalem Mihret, **Lingjun Liu** [EnCyCriS]
  International Workshop on Engineering and Cybersecurity of Critical Systems, 2021.
- **Platooning LEGOs: An Open Physical Exemplar for Engineering Self-Adaptive Cyber-Physical Systems-of-Systems**
  Yong-Jun Shin, **Lingjun Liu**, Sangwon Hyun, Doo-Hwan Bae [SEAMS]
  International Symposium on Software Engineering for Adaptive and Self-Managing Systems, 2021
  Best Artifact Award 🏆
- **MuGenFBD: Automated Mutant Generator for Function Block Diagram Program** [KTSDE]
  Lingjun Liu, Eunkyoung Jee, Doo-Hwan Bae
  KIPS Transactions on Software and Data Engineering, 2021
  – Extended version of the KCSE 2020 paper
- **A Systematic Translation from PAT-based Counterexamples to Viable Test Cases**
  Zelalem Mihret, **Lingjun Liu**, Eunkyoung Jee, Doo-Hwan Bae
  Korea Conference on Software Engineering (KCSE), 2021
- **Analysis of Coupling Effect Hypothesis for Function Block Diagram Programs**
  Lingjun Liu, Eunkyoung Jee, Doo-Hwan Bae
  Korea Software Congress (KSC), 2020
- **Automated Mutant Generation for Function Block Diagram Programs**
  Lingjun Liu, Eunkyoung Jee, Doo-Hwan Bae
  Korea Conference on Software Engineering (KCSE), 2020
  Outstanding Short Paper Award 🏆

RESEARCH EXPERIENCE

North Carolina State University Raleigh, NC, USA
Graduate Research Assistant Aug 2024 – Present

- (Research Project) Causal Evaluation of Structured Context for AI Coding Agents
  - Developing causal evaluation methods for structured AI coding-agent context (AGENTS.md, agent skills), grounded in a systematic review of 50+ papers
- (Research Project) IssueMut: Learning Compiler Fuzzing Mutators from Historical Bugs
  - Built a Python data pipeline that mined 1,760 historical GCC/LLVM bug reports via the GitHub API and HTML parsing into structured test-case pairs, powering IssueMut, a bug-detection framework for C compilers
  - Designed and deployed an agentic LLM workflow using LangChain and Gemini – a multi-agent pipeline with generation, validation, and iterative-revision agents – to synthesize 587 fuzzing mutators from mined test cases; found 65 bugs, 60 of which were confirmed or fixed by GCC/LLVM maintainers
  - Integrated mined mutators into two leading mutational fuzzers (MetaMut and Kitten); augmented variants discovered up to 1.9x more unique crashes than their baselines over 24-hour fuzzing campaigns
  - Mentored undergraduate researchers on mutator scripting and compiler bug reporting, accelerating expansion of the mutator library and shortening the time from bug discovery to reporting
- (Research Project) Differential Testing for Deep Learning Compilers
  - Co-led a differential-testing pipeline for deep learning compilers, parallelized via SLURM job scheduling to run multiple test campaigns concurrently; found 76 bugs across PyTorch, JAX, TensorFlow, and XLA, 45 of which were confirmed or fixed

KAIST Daejeon, South Korea
Full-time Researcher Nov 2023 – Apr 2024 | Oct 2021 – Aug 2022

- (Research Project) Reliability Estimation of Nuclear PLC Software
  Funded by the Korea Foundation of Nuclear Safety (KoFoNS)
  - Built the core reliability computation engine using Bayesian Belief Network modeling for nuclear safety-critical software – developed an initial R/WinBUGS/Shiny prototype, then migrated it to PyMC – to enable regulatory organizations to evaluate nuclear PLC software reliability
  - Designed calculation logic to determine the number of tests required to achieve target reliability thresholds based on pass/fail outcomes
  - Containerized a full-stack application (Next.js, FastAPI, MySQL) with Docker and Docker Compose, and built a one-command deployment script to automate multi-container builds

KAIST Daejeon, South Korea
Graduate Research Assistant Aug 2018 – Aug 2021

- (Research Project) Mutation-Based Test Generation for FBD Programs
  Funded by the National Research Foundation of Korea (NRF) 2019 – 2021
  - Developed MuFBDTester, an automated test generator for safety-critical nuclear reactor protection system software using mutation testing and SMT constraint solving (Yices)
  - Modeled FBD programs and their mutants as SMT constraints to enable systematic test case generation
  - Achieved 100% detection of real industrial faults, reduced manual test-suite creation from days to minutes, and ran 20x faster than leading structural-coverage tools
- (Research Project) [CybWin] Security Testing of Air Traffic Control Systems
  Funded by The Research Council of Norway 2020 – 2021
  - Modeled multi-agent attack scenarios on air traffic control systems using the JADE multi-agent framework
  - Implemented test case generation via model checking to systematically verify system behaviors under attack
- (Research Project) [SW Starlab] Runtime Verification of System-of-Systems
  Funded by the Institute for Information & Communications Technology Promotion (IITP), Korea 2020 – 2021
  - Designed runtime verification property patterns and scopes
  - Developed a verification checker for Mass Casualty Incident-Response (MCI-R) systems to ensure safety properties during operation
- (Research Project) Failure Pattern Analysis for Transportation Systems 2021
  - Applied sequence pattern mining and time-series clustering to detect interaction failures in transportation systems

INDUSTRY EXPERIENCE

Suresoft Technologies Inc. Seongnam, South Korea
Associate Research Engineer Nov 2022 – Jul 2023

- Static Analysis Product Development
  - Shipped production C++ rule checkers for the AUTOSAR C++14 coding guidelines, expanding a 200+ rule suite as a core member of a 3-person team that later grew to 5–7 engineers; advanced from new hire to core contributor within 2 months and increased rule-checker throughput 3x, from 1–2 per week to 5 per week
  - Built a backend license-validation service with Java Spring that enforced time-limited licensing for commercial customers, ensuring compliance and preventing unauthorized use
  - Designed and built a rule-description generator using Java Spring and MongoDB, exposing a REST API for frontend data retrieval and creating the database schema to streamline access to rule metadata for internal tools
  - Managed software packaging with Advanced Installer and collaborated with QA to debug packaging-related test failures
  - Onboarded and mentored two new engineers through pair-programming and code-preview sessions
  - Documented edge cases and checker logic in Jira for code review; supported regression testing for web-based product releases

AWARDS & FELLOWSHIPS

- 2024 University Graduate Fellowship at NCSU – Top graduate student fellowship
- 2024 Best Short Paper Award at the Korea Conference on Software Engineering (KCSE)
- 2021 Best Artifact Award at the International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS)
- 2020 Outstanding Short Paper Award at the Korea Conference on Software Engineering (KCSE)
- 2019 KAIST Scholarship – Merit-based full support

PRESENTATIONS

MuFBDTester: A Mutation-Based Test Sequence Generator for FBD Programs Implementing Nuclear Power Plant Software – Link

- ISSRE – Journal First, Conference Second (J1C2) Track — 2023
- KCSE – Invited talk of excellent international conference/journal papers — 2023

PROFESSIONAL SERVICE

- Student Volunteer, International Symposium on Software Testing and Analysis (ISSTA) — 2027 (upcoming)
- Shadow Program Committee Member, International Conference on Software Engineering (ICSE) — 2027 (ongoing)
- Reviewer, *The Journal of Supercomputing* — 2024
- Student Volunteer, International Conference on Software Engineering (ICSE) — 2020
- Member, Association for Computing Machinery (ACM)
- Member, Special Interest Group on Software Engineering (SIGSOFT)

LEADERSHIP

- Vice President, NCSU Taiwanese Student Association — 2025 – 2026
- Graduate Mentor, Research Experience for Undergraduates (REU) program — 2025

SKILLS

Languages: Python, C++, Java, R
AI-Assisted Development: Claude Code, VS Code, agentic coding workflows, LangChain, Gemini
Systems & Infrastructure: Linux, Docker, SLURM, Git
CI/CD & Testing: Jenkins, GitHub Actions, mutation testing, differential testing, fuzzing, static analysis, SMT solvers (Yices, Z3), regression testing
Frameworks & Tools: REST API, Java Spring, MongoDB, Advanced Installer